

OLIST E-COMMERCE ANALYTICS

PROJECT REPORT

PROJECT DETAIL	DESCRIPTION
Project	Olist E-commerce Analytics
Tool	Microsoft Power BI
Core Technologies	Power Query • DAX • SQL
Dashboard Pages	9 analytical pages
Focus Areas	Sales • Categories • Customers • Delivery • Reviews • Payments • Sellers

The objective of this project was to turn multi-table Olist e-commerce data into an interactive business dashboard that can be used to understand commercial performance, customer behavior and operational quality.

1. Executive Summary

I developed an end-to-end e-commerce analytics solution using the Olist Brazilian e-commerce dataset. The work brings together transaction, customer, product, seller, payment, review and delivery information into a Power BI reporting model.

The final dashboard is designed to move from an executive-level summary to detailed investigation. Users can first understand overall business performance and then explore sales trends, categories, customer geography, logistics, reviews, payment behavior, seller contribution and category-level drillthrough.

The Executive Overview reports 14.21M total revenue, 98.67K orders, 95K customers, 144.01 AOV, 93.23% on-time delivery and a 4.03 average rating.

What I worked on

- Reviewed the structure and purpose of the available Olist datasets.
- Prepared the data for analytical reporting and brought relevant business attributes into the reporting model.
- Created reusable DAX measures for core KPIs and time-based analysis.
- Designed a nine-page Power BI dashboard around practical business questions.
- Analyzed the dashboard outputs and converted them into business-oriented findings.

2. Project Objectives

- Build a structured analytical model from multiple Olist datasets.
- Create reliable KPIs for revenue, orders, customers, AOV and delivery performance.
- Understand category and geographic sales patterns.
- Evaluate customer reviews and the relationship between delivery and satisfaction.
- Analyze payment methods, installments and seller contribution.
- Provide interactive drillthrough capability for deeper category analysis.

3. Dataset & Source Structure

The supplied SQL schema contains nine logical datasets covering customers, geolocation, products, sellers, orders, order items, payments, reviews and product-category translation. The schema includes primary business identifiers such as customer_id, product_id, seller_id and order_id.

DATASET	WHAT IT CONTRIBUTES TO THE PROJECT
Customers	Customer ID, unique customer ID, city and state
Geolocation	ZIP-level latitude, longitude, city and state
Products	Product category and product attributes
Sellers	Seller identity, city and state
Orders	Order status and order lifecycle timestamps
Order Items	Product, seller, price and freight transaction details
Payments	Payment type, installments and payment value
Reviews	Review score, comments and review timestamps
Category Translation	English category names for reporting

4. Data Preparation & Analytical Model

The project uses the analytical fact table “public bi_fact_sales” as the main reporting table, with “DimDate” supporting time-based calculations. The DAX measures supplied for the project reference these model objects.

- Reviewed fields and table structures before building the analytical view.
- Combined the transaction-level information needed for sales, payment, review and delivery analysis.
- Used category translation to make product-category reporting easier to interpret.
- Prepared delivery-related fields such as delivery_days and is_late for logistics KPIs.

- Used a dedicated date table for YTD, previous-year and rolling 30-day analysis.
- Created reusable measures so dashboard visuals respond dynamically to filters.

5. Power BI Dashboard Explanation

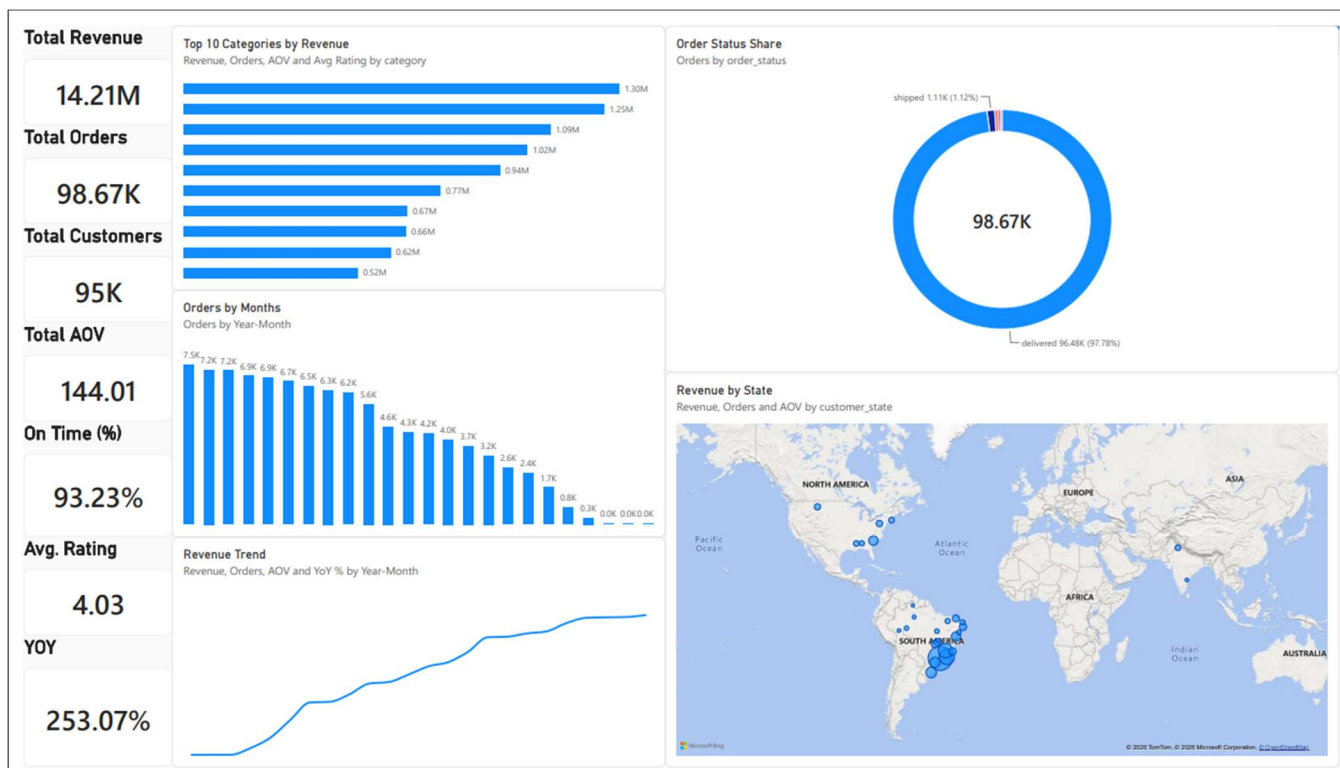
This section explains exactly what is available on each dashboard page, what the visuals are intended to show, and what business questions I used the page to answer.

5.1 Executive Overview

What is on this page:

- KPI cards: Total Revenue, Total Orders, Total Customers, AOV, On-time %, Average Rating and YoY %.
- Revenue Trend showing Revenue, Orders, AOV and YoY % over Year-Month.
- Orders by Month to show order-volume movement over time.
- Top 10 Categories by Revenue for category contribution comparison.
- Revenue by State for geographic performance.
- Order Status Share to show the distribution of order statuses.

Business purpose: Used as the starting point for management-level performance review. It gives a quick view of commercial scale, growth, customer base, service quality and order status. The dashboard shows 14.21M revenue, 98.67K orders, 95K customers, 144.01 AOV, 93.23% on-time and 4.03 average rating.

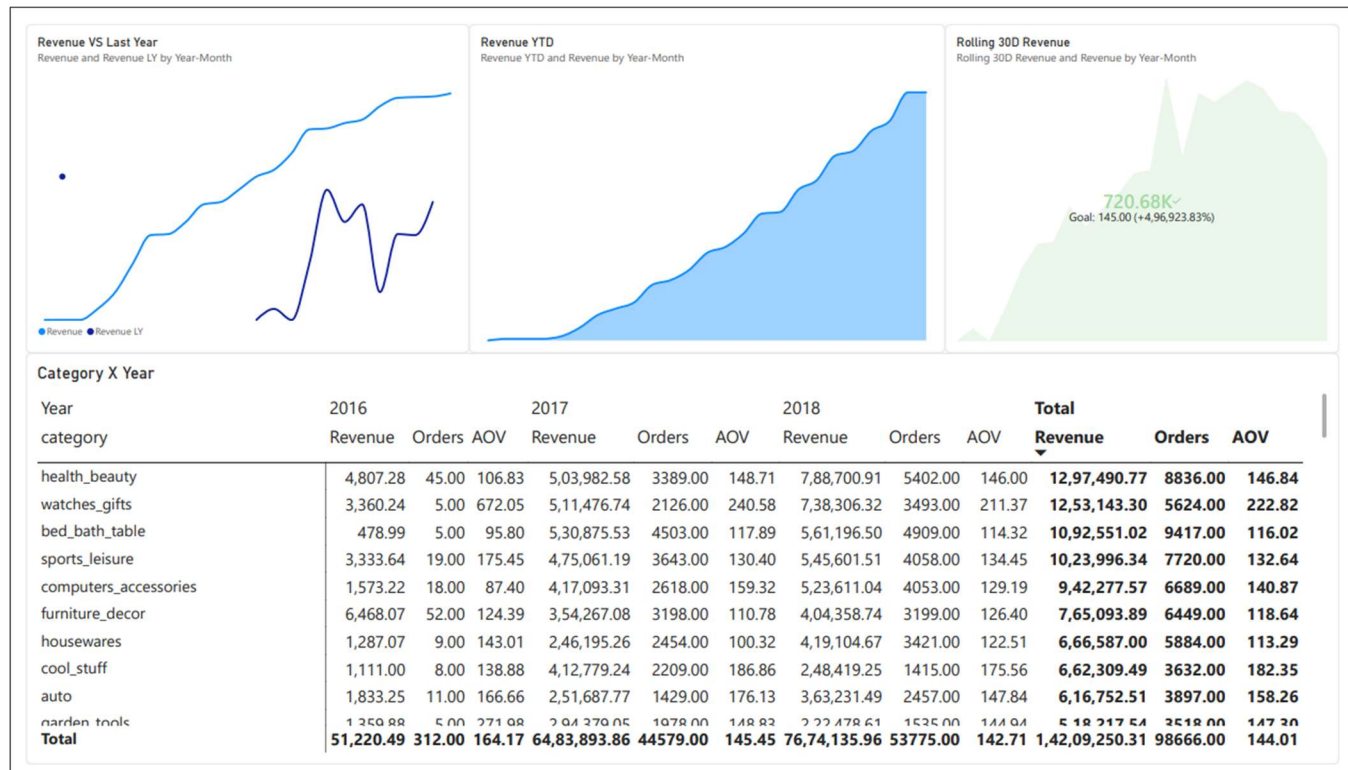


5.2 Sales Trends

What is on this page:

- Revenue vs Last Year comparing current revenue with the previous-year period.
- Rolling 30D Revenue to monitor recent sales momentum.
- Revenue YTD for cumulative year performance.
- Category × Year table comparing Revenue, Orders and AOV across 2016, 2017 and 2018.

Business purpose: Used to understand growth, seasonality and changes in category performance over time. The page specifically uses the project's time-intelligence measures such as Revenue LY, YoY %, Revenue YTD and Rolling 30D Revenue.

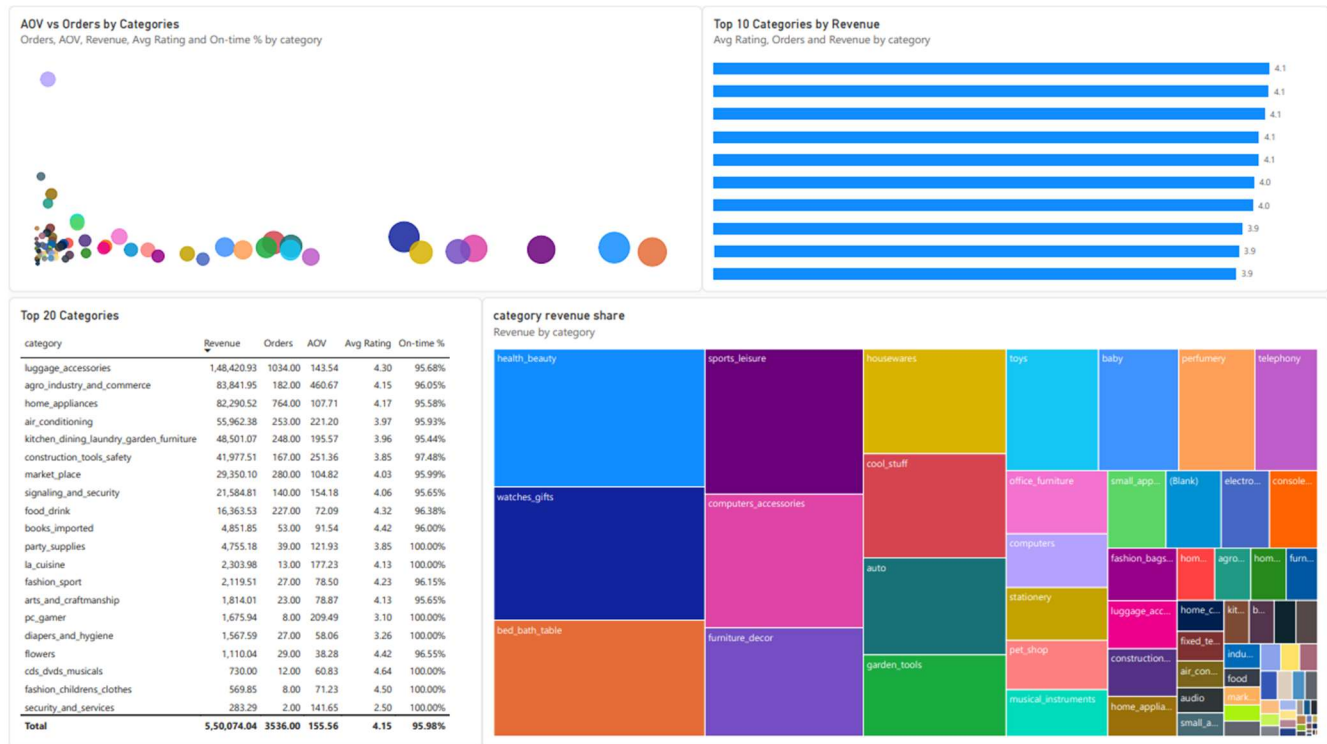


5.3 Category & Products

What is on this page:

- Category revenue share showing how revenue is distributed across categories.
- AOV vs Orders by Categories to compare order volume with average order value.
- Top 10 Categories by Revenue.
- Top 20 category table containing Revenue, Orders, AOV, Avg Rating and On-time %.

Business purpose: Used to identify high-contribution categories and compare commercial performance with customer experience and delivery quality. For example, the category table includes luggage_accessories with 148,420.93 revenue, 1,034 orders and 4.30 average rating.

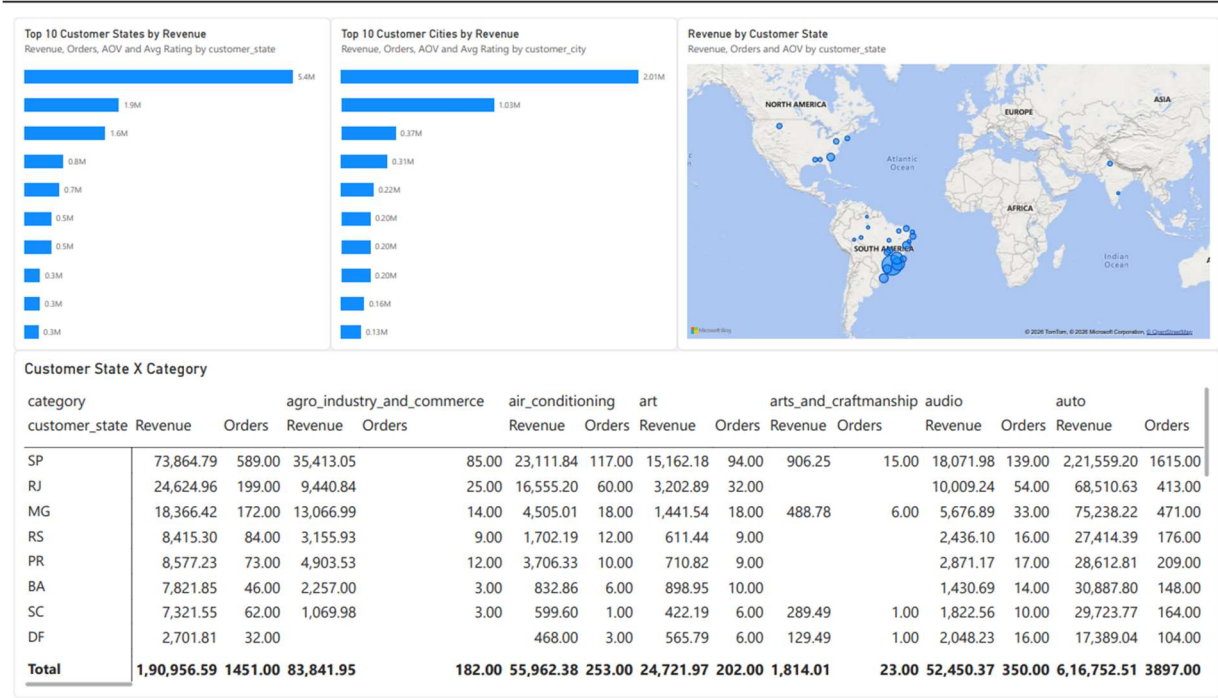


5.4 Customer Geographics

What is on this page:

- Revenue by Customer State.
- Top 10 Customer States by Revenue.
- Customer State × Category analysis.
- Top 10 Customer Cities by Revenue.

Business purpose: Used to understand where customers and revenue are concentrated geographically and to compare category demand across locations.

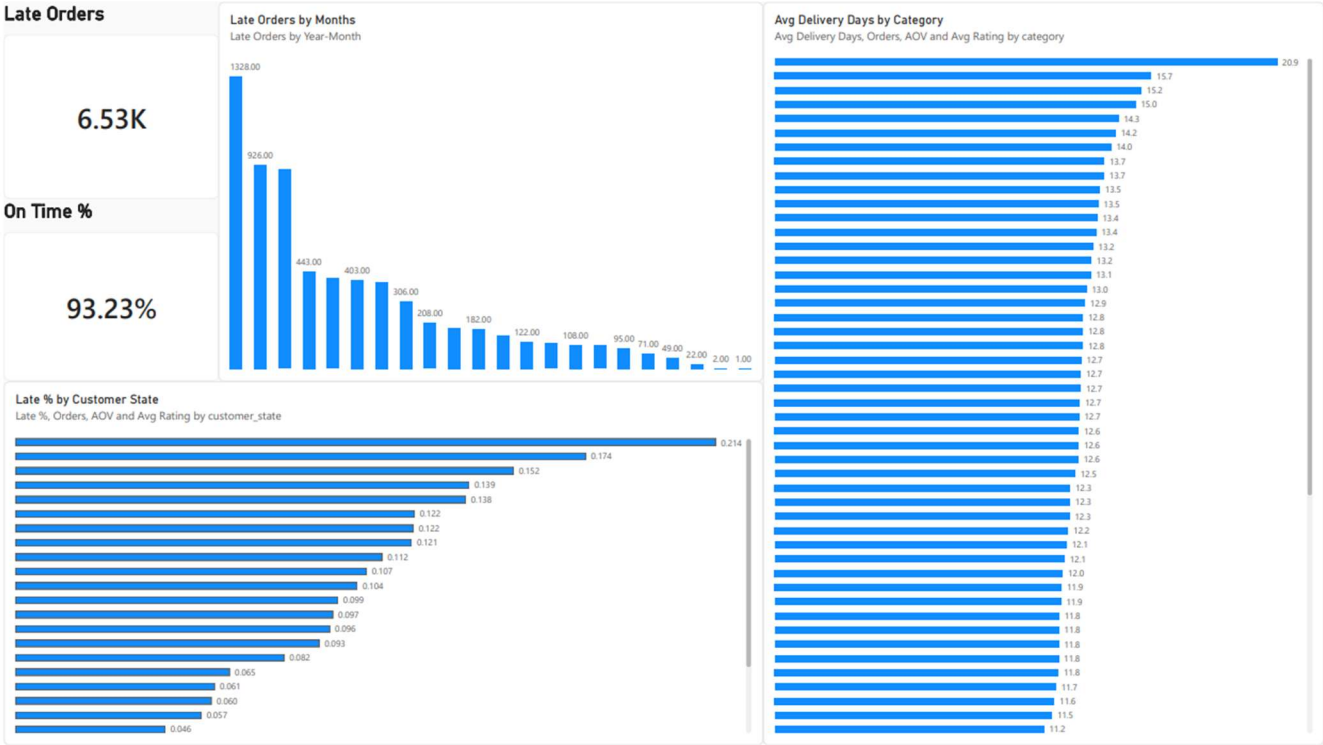


5.5 Delivery & Logistics

What is on this page:

- On-time % KPI.
- Late Orders by Month / Year-Month.
- Average Delivery Days by Category.
- Late % by Customer State.

Business purpose: Used to monitor service quality and identify where delivery performance needs attention. The page reports 93.23% on-time delivery and approximately 6.53K late orders.

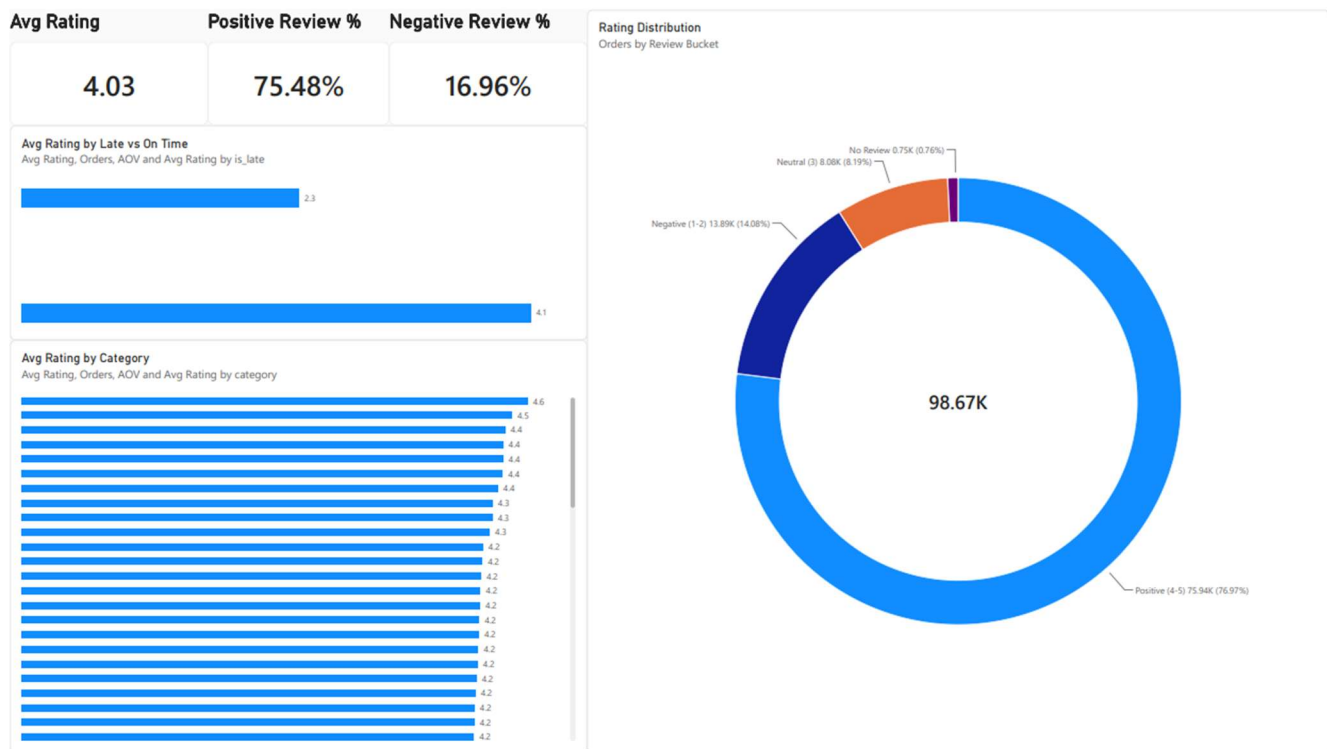


5.6 Reviews

What is on this page:

- Average Rating KPI.
- Positive Review % and Negative Review %.
- Rating Distribution.
- Orders by Review Bucket: Positive, Negative, Neutral and No Review.
- Average Rating by Late vs On Time.
- Average Rating by Category.

Business purpose: Used to understand customer satisfaction and connect reviews with delivery performance. The dashboard shows 4.03 average rating, 75.48% positive reviews and 16.96% negative reviews. It also shows average rating of 4.1 for on-time versus 2.3 for late orders.

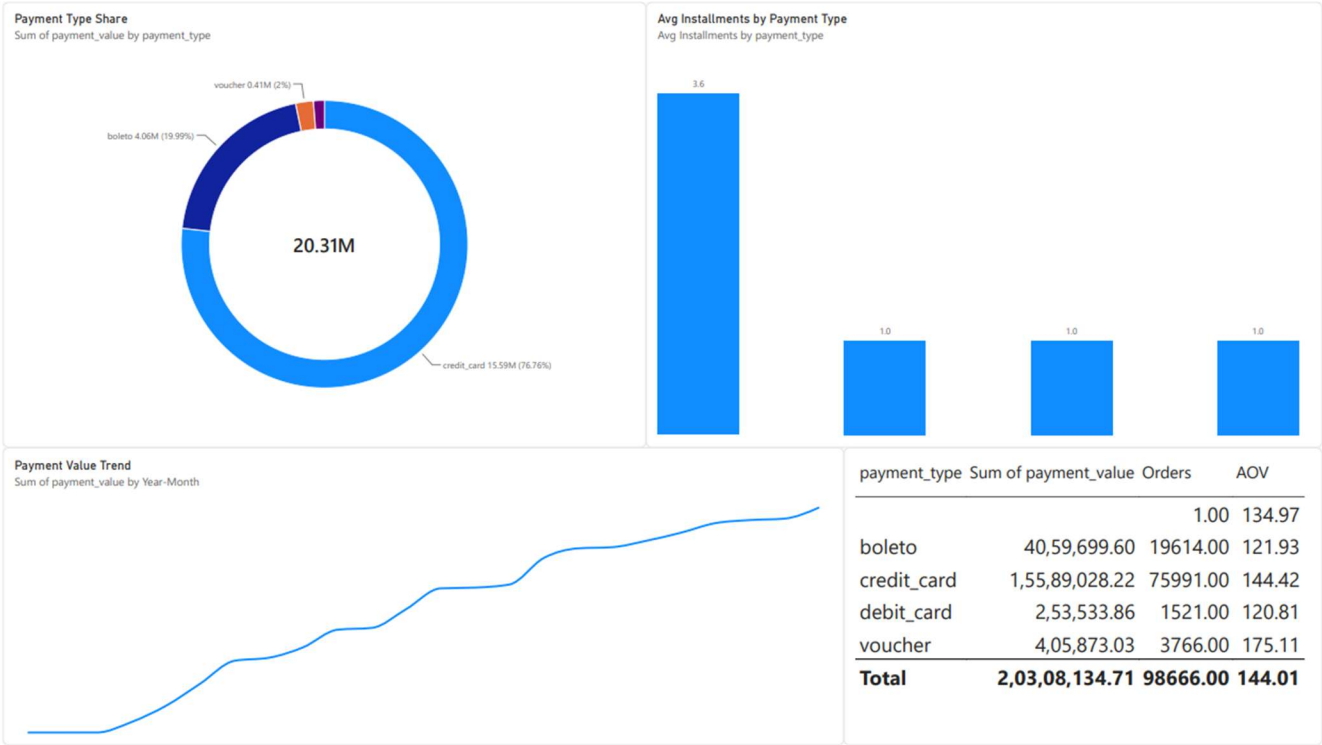


5.7 Payments

What is on this page:

- Payment Type Share.
- Average Installments by Payment Type.
- Payment Value Trend by Year-Month and payment type.
- Payment Value, Orders and AOV by payment type.

Business purpose: Used to understand customer payment preferences and installment behavior. Credit card represents 76.76% of payment value and boleto 19.99% in the dashboard.

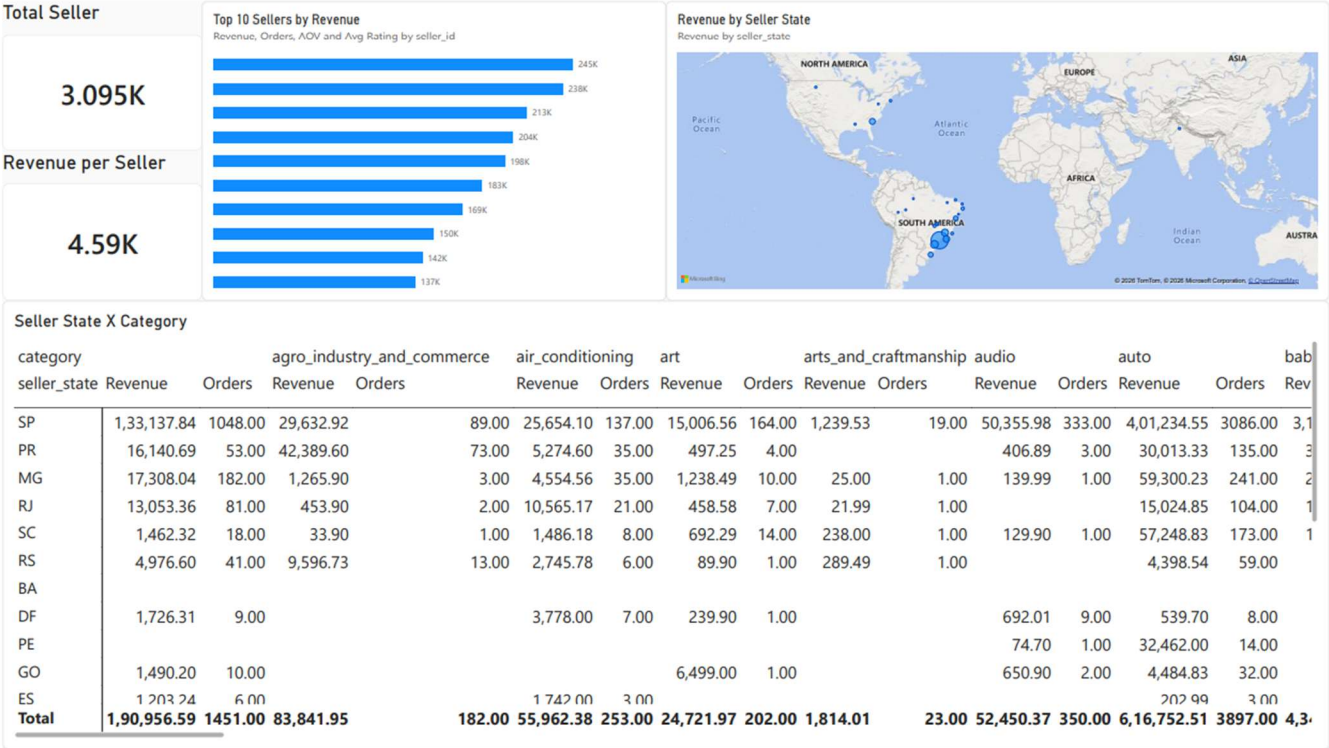


5.8 Sellers

What is on this page:

- Total Seller KPI.
- Revenue per Seller KPI.
- Revenue by Seller State.
- Top 10 Sellers by Revenue.
- Seller State × Category analysis.

Business purpose: Used to compare seller contribution, identify high-performing sellers and understand seller-side geographic/category patterns. The dashboard reports 3.095K sellers and 4.59K revenue per seller.



Seller State X Category

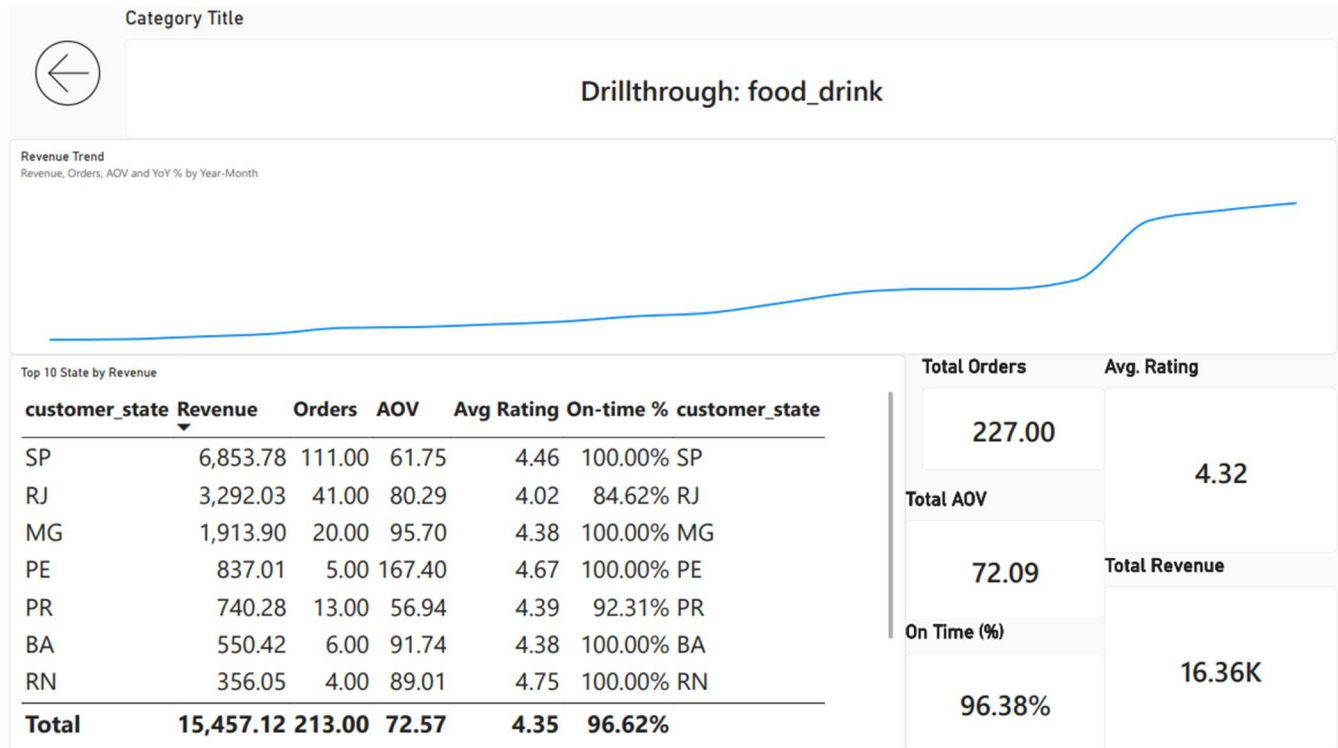
category	agro_industry_and_commerce				air_conditioning		art	arts_and_craftmanship		audio	auto		bab	
seller_state	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders
SP	1,33,137.84	1048.00	29,632.92		89.00	25,654.10	137.00	15,006.56	164.00	1,239.53	19.00	50,355.98	333.00	4,01,234.55
PR	16,140.69	53.00	42,389.60		73.00	5,274.60	35.00	497.25	4.00			406.89	3.00	30,013.33
MG	17,308.04	182.00	1,265.90		3.00	4,554.56	35.00	1,238.49	10.00	25.00	1.00	139.99	1.00	59,300.23
RJ	13,053.36	81.00	453.90		2.00	10,565.17	21.00	458.58	7.00	21.99	1.00			15,024.85
SC	1,462.32	18.00	33.90		1.00	1,486.18	8.00	692.29	14.00	238.00	1.00	129.90	1.00	57,248.83
RS	4,976.60	41.00	9,596.73		13.00	2,745.78	6.00	89.90	1.00	289.49	1.00			4,398.54
BA														
DF	1,726.31	9.00				3,778.00	7.00	239.90	1.00			692.01	9.00	539.70
PE												74.70	1.00	32,462.00
GO	1,490.20	10.00						6,499.00	1.00			650.90	2.00	4,484.83
ES	1,203.24	6.00				1,742.00	3.00							202.99
Total	1,90,956.59	1451.00	83,841.95		182.00	55,962.38	253.00	24,721.97	202.00	1,814.01	23.00	52,450.37	350.00	6,16,752.51

5.9 Drill through

What is on this page:

- Dynamic drillthrough title based on selected category.
- Category-level Revenue, Orders, AOV, On-time % and Avg Rating.
- Revenue trend for the selected category.
- Top states by revenue with Orders, AOV, Avg Rating and On-time %.

Business purpose: Used for detailed investigation after identifying an interesting category on the main dashboard. For the displayed food_drink example, the page shows 16.36K revenue, 227 orders, 72.09 AOV, 96.38% on-time and 4.32 average rating.



6. DAX & KPI Development

I created reusable DAX measures rather than depending only on raw columns. The measure set covers base sales KPIs, time intelligence, review sentiment, payments, delivery, seller performance and drillthrough. The supplied DAX list contains the formulas for these measures.

MEASURE GROUP	EXAMPLES	PURPOSE
Base Measures	Revenue, Orders, Customers, AOV, On-time %	Core business KPIs
Time Intelligence	Revenue YTD, Revenue LY, YoY %, Rolling 30D Revenue	Growth and trend analysis
Review Sentiment	Positive Reviews %, Negative Reviews %	Customer sentiment
Payments	Payment Value, Avg Installments	Payment behavior
Delivery	Avg Delivery Days, Late %	Logistics performance
Reviews	Review Bucket	Review segmentation
Seller	Sellers, Revenue per Seller	Seller contribution
Drillthrough	Drill Title	Interactive category navigation

Important DAX functions used include SUM, DISTINCTCOUNT, COUNTROWS, DIVIDE, CALCULATE, AVERAGE, TOTALYTD, SAMEPERIODLASTYEAR, DATESINPERIOD, SWITCH, SELECTEDVALUE and COALESCE.

7. Key Findings & Insights

Strong overall scale: The dashboard reports 14.21M revenue, 98.67K orders and 95K customers, giving a clear picture of the overall scale of the analyzed business.

Delivery is generally strong, but late orders matter: Overall on-time delivery is 93.23%, while the dashboard records about 6.53K late orders. This indicates that delivery is broadly reliable but still represents a meaningful operational improvement area.

Customer satisfaction is mostly positive: Average rating is 4.03 and the dashboard reports 75.48% positive reviews against 16.96% negative reviews.

Delivery quality is strongly connected with reviews: The average rating is 4.1 for on-time orders compared with 2.3 for late orders. This makes delivery performance an important customer-experience metric.

Credit card dominates payments: Credit card contributes 76.76% of payment value, while boleto contributes 19.99%. Credit-card transactions also show higher average installments.

Category performance differs substantially: The category-by-year analysis shows meaningful differences in revenue, order volume and AOV across categories, supporting category-specific decision making.

8. Challenges Encountered

- Working with multiple related datasets required careful use of common business identifiers such as order_id, customer_id, product_id and seller_id.
- The geolocation dataset contains repeated ZIP-code prefixes, so it cannot be treated as a simple one-row-per-ZIP dimension.
- Time-based calculations required a consistent date table and appropriate DAX time-intelligence logic.
- Delivery analysis depended on correctly prepared delivery duration and late-status fields.
- Large transaction-level data required reusable measures and a structured reporting model to keep analysis manageable.

9. Limitations

- The analysis is limited to the fields and definitions available in the supplied Olist model.

- Dashboard values should be interpreted according to the project's existing metric definitions.
- The report focuses on descriptive and diagnostic analytics; it does not claim predictive forecasting or causal relationships.
- Geographic and operational interpretations should be validated against the underlying business context before making production decisions.

10. Conclusion

This project demonstrates my practical approach to an end-to-end analytics problem: starting with multiple raw business datasets, understanding their structure, preparing them for reporting, building reusable DAX calculations, designing a multi-page Power BI dashboard and converting the results into business insights.

The final solution brings commercial and operational perspectives together in one place. It can be used to review sales performance, category contribution, customer geography, delivery reliability, customer satisfaction, payment behavior and seller performance, while the drillthrough page provides a path from summary-level analysis to category-level investigation.

Through this project, I applied practical skills in Power BI, data preparation, DAX, SQL-based data modeling, KPI design and business-oriented data storytelling.

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